NAS Cleanup: High-Performance File Deduplication and ML-Powered Organisation for Synology NAS

BLAKE3 Hashing, BTRFS Reflink Dedup, and Neural Image Classification

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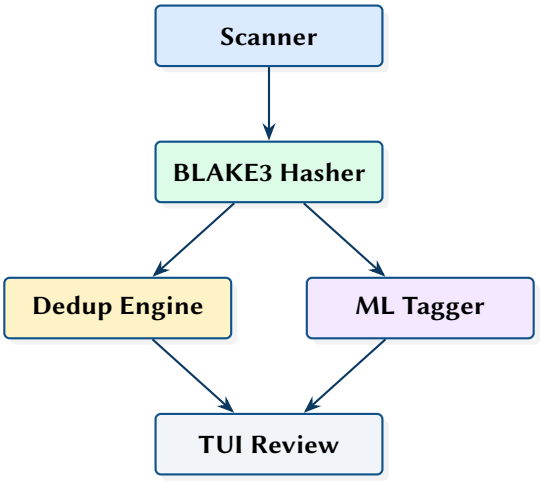

Abstract. This report presents a Rust application for managing large-scale file collections on Synology NAS systems. The tool traverses millions of files using parallel directory walking (jwalk), identifies bit-for-bit duplicates via BLAKE3 hashing with partial-hash pre-filtering, and deduplicates using BTRFS reflinks (copy-on-write) to reclaim space without deleting data. An ML classification pipeline using candle (MobileNetV3 and ResNet50) automatically categorises images and writes tags to XMP sidecar files. A ratatui-based TUI provides interactive review of duplicates and organisation plans. The tool is optimised for astrophotography and conventional photography workflows on Synology DSM.

Version	Date	Changes
0.1.0	March 2026	Initial architecture report

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1 Introduction and Motivation

Large Network Attached Storage (NAS) systems accumulate duplicate files over years of use, particularly in photography workflows where RAW files are copied between editing workstations, imported from multiple cards, and backed up across volumes. Traditional deduplication tools are either too slow for million-file collections or require destructive deletion.

NAS Cleanup takes a different approach: it uses B-tree File System (BTRFS) reflinks to deduplicate at the block level, replacing duplicate files with Copy-on-Write (CoW) clones that share physical storage while maintaining independent file entries. This is non-destructive—each file retains its own path and metadata.

: Safety First

All destructive operations (delete, reflink, organise) support mandatory dry-run mode. The TUI defaults to read-only preview. The organise command generates a rollback journal that can undo all file moves.

2 Scanning and Hashing

2.1 Parallel Traversal

Directory traversal uses jwalk [1] for parallel walking with Synology-specific exclusion patterns (`@eaDir`, `#recycle`, `.DS_Store`).

2.2 BLAKE3 Hashing

File hashing uses BLAKE3 [2] for SIMD-accelerated content hashing. A two-phase approach minimises disk I/O:

1. **Head hash** — Hash the first 16 KB of each file. Files with unique head hashes are immediately classified as unique.
2. **Full hash** — Only files with matching head hashes proceed to full-content hashing.

```
pub fn hash_file(path: &Path, partial: bool) -> Result<Hash> {
    let mut hasher = blake3::Hasher::new();
    let file = File::open(path)?;
    let mut reader = BufReader::with_capacity(65536, file);
    let limit = if partial { 16384 } else { usize::MAX };
    let mut total = 0;
    loop {
        let buf = reader.fill_buf()?;
        if buf.is_empty() || total >= limit { break; }
        let n = buf.len().min(limit - total);
        hasher.update(&buf[..n]);
        reader.consume(n);
        total += n;
    }
    Ok(hasher.finalize())
}
```

3 Duplicate Detection Pipeline

The four-stage pipeline progressively filters candidates:

1. **Discovery** — Parallel traversal builds file index

Insight: The partial-hash optimisation reduces disk reads by 90%+ on typical photo collections where most files have unique first 16 KB (different EXIF timestamps, camera serial numbers, frame counters). Only genuine duplicates require full-file reads.

2. **Size filter** — Group by file size; unique sizes are immediately unique
3. **Head-hash filter** — 16 KB partial hash within size groups
4. **Full-hash verification** — Complete BLAKE3 hash for remaining candidates

Results are stored in a sled [3] embedded database for persistent indexing across sessions.

3.1 BTRFS Reflink Deduplication

On BTRFS volumes, duplicates are deduplicated using the `FICLONE` ioctl, which replaces file data with a CoW reference to the original. The `doctor` command verifies reflink support before operations.

> **Design Decision 1: Reflink vs Delete**
>
> Reflink deduplication is preferred over deletion because it preserves all file paths and metadata. Applications that reference files by path continue to work. The storage saving is equivalent to deletion, but the operation is reversible by simply copying (which triggers CoW, allocating new blocks).

4 ML Image Classification

The ML pipeline uses candle [4] for inference:

Table 1. ML classification pipeline.

Pass	Model	Purpose
Fast scan	MobileNetV3	Basic categories (Landscape, Portrait, Night Sky, Architecture, etc.)
Detailed	ResNet50	Fine-grained taxonomy with confidence scores

Classification results are written to XMP sidecar files (`.xmp`) alongside the original images, preserving tags in a standard format readable by Lightroom, Capture One, and other photo editors.

5 TUI Review Interface

The ratatui [5] TUI provides interactive review of duplicates with:

- Side-by-side file metadata comparison
- Bulk selection with keyboard shortcuts
- Dry-run preview of all operations
- Real-time progress during scan and hash operations

Insight: *Embedded JPEG extraction from RAW files (CR2, NEF, ARW, DNG) avoids the overhead of full RAW decoding for classification. The embedded preview is sufficient for scene classification and is typically 10–100× faster to extract than full demosaicing.*

6 Design Summary

Table 2. Key design decisions.

Decision	Rationale
BLAKE3	SIMD-accelerated; 3–5× faster than SHA-256
Partial hash pre-filter	90%+ disk read reduction on photo collections
BTRFS reflink	Non-destructive dedup; preserves all file paths
sled database	Persistent index across sessions; crash-safe
Two-pass ML	Fast MobileNetV3 triage; detailed ResNet50 only when needed
Embedded JPEG extraction	Avoids RAW decode overhead for classification

References

[1] jwalk contributors, *Jwalk: Blazing fast filesystem walk*, 2024. [Online]. Available: https://crates.io/crates/jwalk

[2] J. O'Connor et al., *BLAKE3: One function, fast everywhere*, 2024. [Online]. Available: https://github.com/BLAKE3-team/BLAKE3

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[4] Hugging Face, *Candle: Minimalist ML framework for Rust*, 2024. [Online]. Available: https://github.com/huggingface/candle

[5] Ratatui contributors, *Ratatui: A Rust library for cooking up TUIs*, 2024. [Online]. Available: https://ratatui.rs